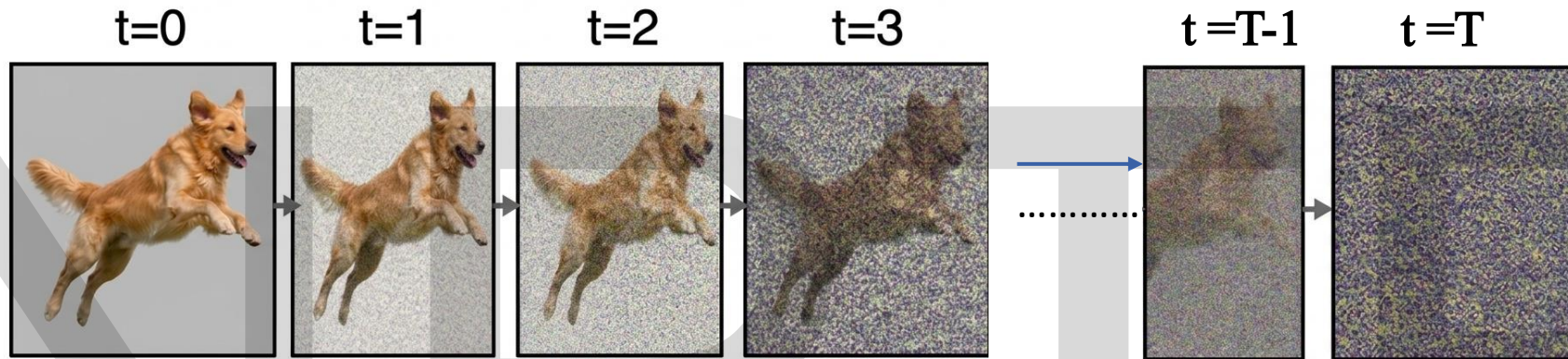


NPTEL

Denoising Diffusion Probabilistic Model -Reverse Process

Recap on Forward Diffusion Process



$$q(x_t|x_0) = \mathcal{N}(x_t; \sqrt{\bar{\alpha}_t} x_0; (1 - \bar{\alpha}_t)I)$$

$$x_t = \sqrt{\bar{\alpha}_t} x_0 + \sqrt{(1 - \bar{\alpha}_t)} \epsilon$$

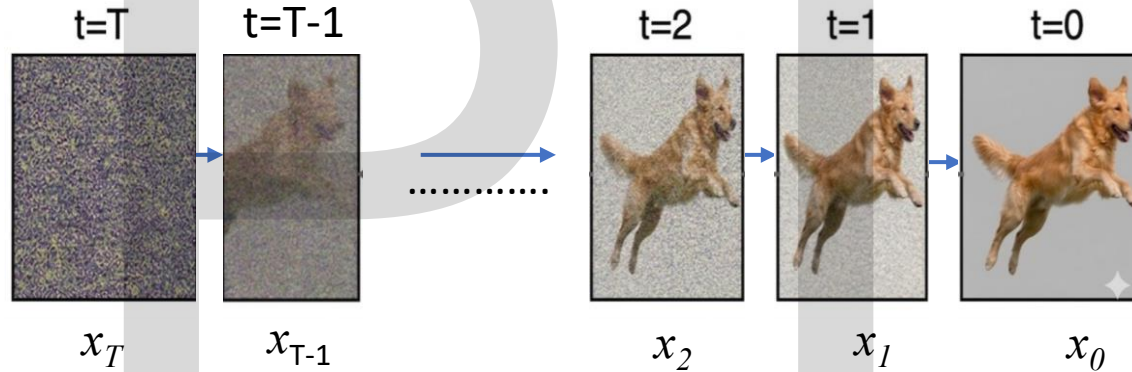
Using the reparameterization trick, x_t can be sampled directly from x_0 without performing all intermediate diffusion steps.

Backward Diffusion Process

Gradually removes
noise

$$x_T \rightarrow x_{T-1} \rightarrow \cdots x_2 \rightarrow x_1 \rightarrow x_0$$

Generate realistic samples by progressively removing
noise from a Gaussian noise sample.



Goal of Backward Process

We want to learn:

$$p_{\theta}(x_{t-1} | x_t)$$

Given a noisy sample x_t , predict a slightly less noisy version x_{t-1}

- ✓ Goal of DDPM training is to learn this distribution
- ✓ Θ : Parameters of the neural network

DDPM assumes the reverse process is a **Markov chain**.

Markov Chain

The reverse process is assumed to be a Markov chain, meaning x_{t-1} depends only on x_t

$$P(x_{t-1}|x_t, x_{t+1}, \dots, x_T) = P(x_{t-1}|x_t)$$

$$x_T \rightarrow \dots \rightarrow x_{t+1} \rightarrow x_t \rightarrow x_{t-1} \dots x_0$$

Because of this Markov assumption:

$$p_{\theta}(X_{0:T}) = p(X_T) \prod_{t=1}^T p_{\theta}(x_{t-1}|x_t)$$

Joint probability
distribution for the
entire reverse trajectory

Prior distribution over X_T
(Starting point of generation)

$$X_T \sim N(0, I)$$

Equations from Forward diffusion Process

$$q(x_t|x_0) = \mathcal{N}(x_t; \sqrt{\bar{\alpha}_t} x_0, (1 - \bar{\alpha}_t)I)$$

→ Probability distribution of Forward Diffusion process

Prior distribution

A probability distribution that represents what we know (or assume) about a variable before observing additional information.

Posterior distribution

A probability distribution after observing additional information.

Posterior of the forward Process

$$q(x_{t-1}|x_t, x_0)$$



Under the forward (noising) process q , what is the probability distribution of the previous state x_{t-1} , given the current noisy state x_t and the original data x_0

Posterior of the forward Process

$$q(x_{t-1}|x_t, x_0) = \mathcal{N}(x_{t-1}, \tilde{\mu}_t(x_t, x_0), \tilde{\beta}I)$$

Mean:

$$\tilde{\mu}_t(x_t, x_0) = \frac{\sqrt{\bar{\alpha}_{t-1}} \beta_t}{1 - \bar{\alpha}_t} x_0 + \frac{\sqrt{\alpha_t}(1 - \bar{\alpha}_{t-1})}{1 - \bar{\alpha}_t} x_t$$

Variance:

$$\tilde{\beta}_t = \frac{1 - \bar{\alpha}_{t-1}}{1 - \bar{\alpha}_t} \beta_t$$

$$\bar{\alpha}_t = \prod_{s=1}^t \alpha_s$$

This is Gaussian and can be computed exactly

Reverse Distribution of Backward Process

$$x_T \rightarrow x_{T-1} \rightarrow \cdots x_2 \rightarrow x_1 \rightarrow x_0$$

$p_\theta(x_{t-1}|x_t)$: probability of x_{t-1} given x_t

Can be modeled as

$$p_\theta(x_{t-1}|x_t) = \mathcal{N}(x_{t-1}; \mu_\theta(x_t, t), \Sigma_\theta(x_t, t))$$

$$x_{t-1} \sim \mathcal{N}(\mu_\theta(x_t, t), \Sigma_\theta(x_t, t))$$

The next denoised image x_{t-1} is sampled from a Gaussian whose mean and variance are predicted from x_t .

The Neural Network must learn the mean and the covariance

Recalling the Mean and variance equation

Mean:

$$\tilde{\mu}_t(x_t, x_0) = \frac{\sqrt{\bar{\alpha}_{t-1}} \beta_t}{1 - \bar{\alpha}_t} x_0 + \frac{\sqrt{\alpha_t}(1 - \bar{\alpha}_{t-1})}{1 - \bar{\alpha}_t} x_t$$

Variance:

$$\tilde{\beta}_t = \frac{1 - \bar{\alpha}_{t-1}}{1 - \bar{\alpha}_t} \beta_t$$

Can we use this equation?

No, because the posterior mean depends on x_0 , which is unknown during generation.

Solution?

Rewrite these equations without x_0

True Reverse Distribution

From forward diffusion

$$x_t = \sqrt{\alpha_t} x_0 + \sqrt{(1 - \alpha_t)} \epsilon$$

Express x_0 Through Noise

$$x_0 = \frac{x_t - \sqrt{(1 - \alpha_t)} \epsilon}{\sqrt{\alpha_t}}$$

Substituting into $\tilde{\mu}_t$:

$$\tilde{\mu}_t = \frac{1}{\sqrt{\alpha_t}} \left(x_t - \frac{\beta_t}{\sqrt{(1 - \alpha_t)}} \epsilon \right)$$

Replace true noise by the noise predicted by the NN

$$\epsilon \approx \epsilon_{\theta}(x_t, t)$$

$$\tilde{\mu}_{\theta}(x_t, t) = \frac{1}{\sqrt{\alpha_t}} \left(x_t - \frac{\beta_t}{\sqrt{(1 - \alpha_t)}} \epsilon_{\theta}(x_t, t) \right)$$

Learned Reverse Distribution

$$p_{\theta}(x_{t-1}|x_t) = \mathcal{N}(\mu_{\theta}(x_t, t), \sigma_t^2 I)$$

Sampling Equation

$$x_{t-1} = \mu_{\theta}(x_t, t) + \sigma_t z$$

$$z \sim \mathcal{N}(0, I)$$

Substitute μ_{θ}

$$x_{t-1} = \frac{1}{\sqrt{\alpha_t}} \left(x_t - \frac{\beta_t}{\sqrt{1-\alpha_t}} \epsilon_{\theta}(x_t, t) \right) + \sigma_t z$$

Training of Diffusion Model

Loss Function:

$$\mathcal{L} = \mathbb{E}_{x_0, \epsilon, t} [\|\epsilon - \epsilon_\theta(x_t, t)\|^2]$$

Mean Squared Error

True Noise

Predicted Noise

Expectation value over x_0 , ϵ and t

Train the model so it performs well on average
across all:

- ✓ images
- ✓ noise levels
- ✓ timesteps

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Summary

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*Next Session:
U-Net architecture for Denoising*